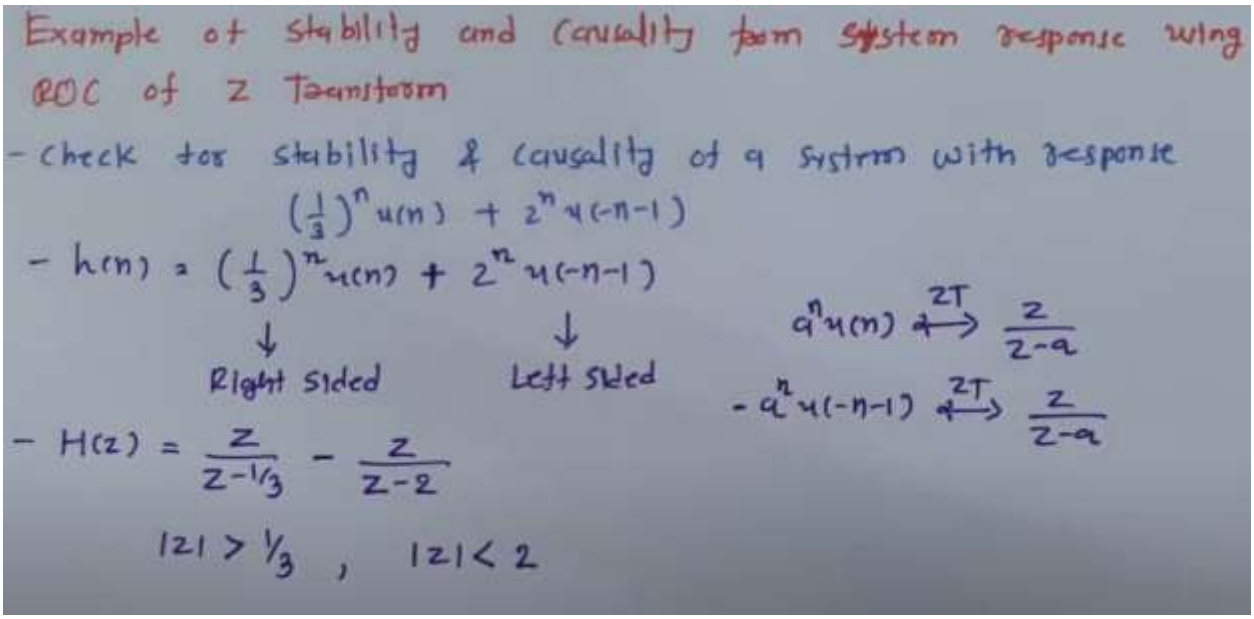
Example of Z-Transform

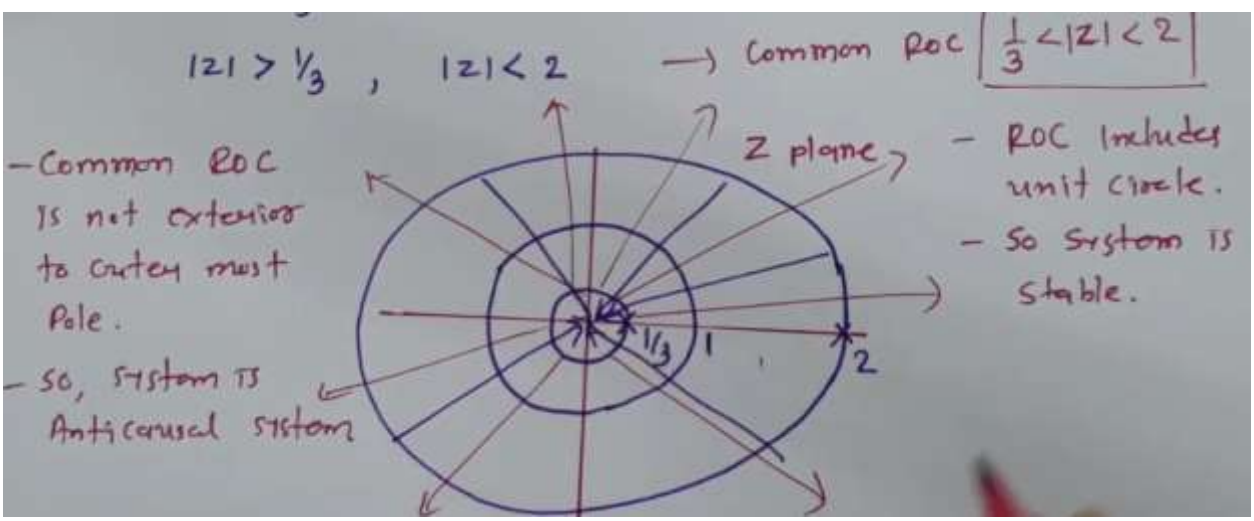
Casual: ROC is outside the outermost pole

Stable: ROC must include unit circle

Both Stable and Casual: ROC must include unit circle and ROC is outside the outermost pole

Example 1:





< - Anticasual

>-Casual

Both – Noncasual

Example 2:

Example of Stability and Causality from System response using ROC of z Transform

- Check for stability & causality of a system with

$$x(n) = \left(\frac{1}{2}\right)^n u(n) + 3^n u(-n-1)$$

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$$= \underbrace{\left(\frac{1}{2}\right)^n u(n)}_{\text{Right Sided}} + \underbrace{\left(\frac{1}{3}\right)^n u(-n-1)}_{\text{Left Sided}}$$

$$a^n u(n) \xleftrightarrow{ZT} \frac{z}{z-a}$$

$$-a^n u(-n-1) \xleftrightarrow{ZT} \frac{z}{z-a}$$

$$X(z) = \underbrace{\frac{z}{z-1/2}}_{|z| > 1/2} - \underbrace{\frac{z}{z-1/3}}_{|z| < 1/3}$$

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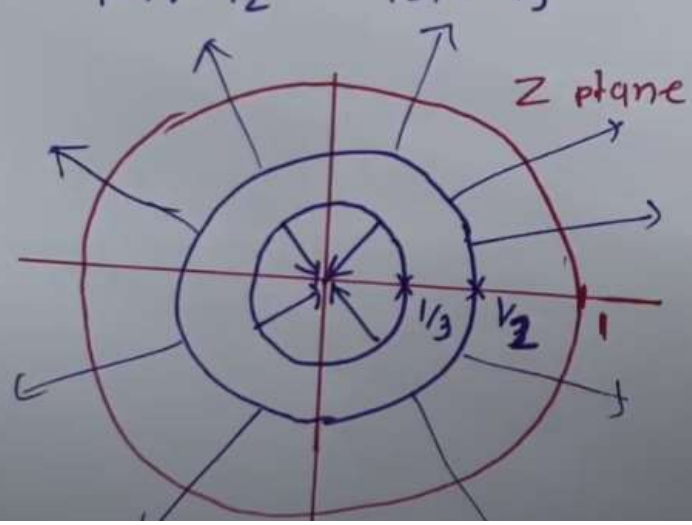

→ Here there is no common ROC.

→ So stability and causality can be justified here.